

## Article

# Study on the Occurrence States and Enrichment Mechanisms of the Dispersed Elements Ga, Ge, and In in the Chipu Pb-Zn Deposit, Sichuan Province, China

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**Abstract:** The dispersed elements Ga, Ge, and In are crucial strategic mineral resources often enriched in Pb-Zn deposits. The Chipu Pb-Zn deposit, located on the western edge of the Yangtze Block, lies to the north of the Sichuan-Yunnan-Guizhou (SYG) Pb-Zn metallogenic province with large amounts of Emeishan basalt. Based on trace element and in situ sulfur isotope analyses by (LA)-ICP-MS, sphalerite is the main carrier mineral for Ga (17~420 ppm), Ge (3.87~444 ppm), and In (31~720 ppm). Ga or Ge correlate significantly with Cu, while In substitutes for Zn in sphalerite alongside Fe. Key substitution reactions include  $\text{Ga}^{3+} + \text{Cu}^+ \leftrightarrow 2\text{Zn}^{2+}$ ,  $\text{Ge}^{4+} + 2\text{Cu}^+ \leftrightarrow 3\text{Zn}^{2+}$ , and  $2\text{In}^{3+} + \text{Fe}^{2+} \leftrightarrow 4\text{Zn}^{2+}$ . Sphalerite crystallized at medium to low temperatures (114–195 °C). Sulfide  $\delta^{34}\text{S}$  values (+3.48 to +24.74‰) suggest sulfur mainly originated from Dengying Formation marine sulfates via thermochemical sulfate reduction (TSR). Metal-bearing fluid release at 30 Ma post-Emeishan mantle plume activity (261–257 Ma) coincides with the Chipu deposit's mineralization period (230–200 Ma), suggesting the Chipu deposit is associated with Emeishan plume activity. The magmatic activity drove basinal brine circulation, extracting In from intermediate-felsic igneous rocks and metamorphic basement. Elevated temperatures promoted the coupling of Fe and In into sphalerite, causing anomalous In enrichment.

**Keywords:** dispersed elements Ga, Ge, and In; LA-ICP-MS; sphalerite; anomalous enrichment; Chipu Pb-Zn deposit



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## 1. Introduction

Gallium (Ga), germanium (Ge), and indium (In), as key dispersed elements with unique characteristics, are vital for the growth of diverse emerging industries [1,2]. Nonetheless, the challenge lies in adequately enriching dispersed elements and forming independent dispersed element deposits (with rare exceptions) due to the low average abundances of these elements in the Earth's crust (mostly at  $10^{-6}$ ~ $10^{-9}$  level) [3–5]. It has been proven that the Pb-Zn deposits are enriched in high-grade dispersed Ga, Ge, and In [6,7]. Studies over the past few decades have shown that the ore minerals found in different Pb-Zn deposits (such as MVT, Skarn, Epithermal, VMS, and SEDEX) contain sufficiently high concentrations of Ga, Ge, and In, primarily carried by sphalerite [8–13].

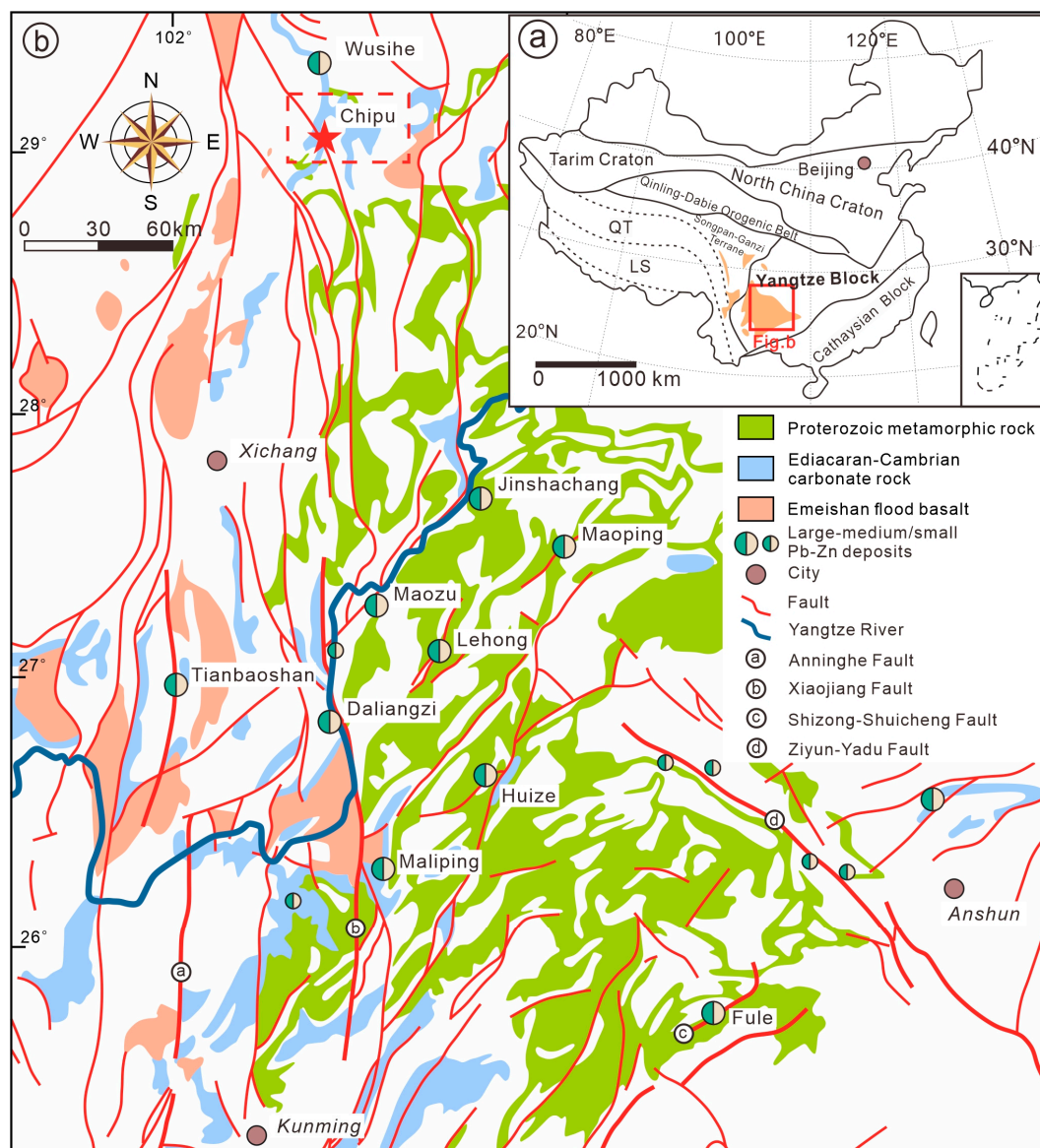
Situated on the southwestern margin of the Yangtze Block, the Sichuan-Yunnan-Guizhou (SYG) Pb-Zn metallogenic province forms a crucial segment of the extensive low-temperature metallogenic domain in Southwest China, encompassing more than 400 Pb-Zn deposits. Reports indicate that numerous ore deposits in the area, which are key

sources of Pb, Zn, and Ag in China [14,15], are also associated with dispersed elements like Ga and Ge (e.g., Tianbaoshan; Niujaotang; Huize; Maliping) [16–20].

The Chipu Pb-Zn deposit is located in the northern part of the SYG Pb-Zn metallogenic province. Earlier research primarily focused on the fundamental geological features, geochronology, isotope geochemistry, and ore-forming fluid in the deposit [21–27]; the potential value of dispersed elements within the deposit has yet to be assessed. Here, we employed electron microprobe analysis (EMPA) and in situ laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) to methodically analyze trace elements and sulfur isotopes in typical sulfides from the deposit. The results suggest an enrichment of Ga and Ge in the Chipu deposit, with an unusual enrichment in In. This research focuses on the occurrence mechanisms of Ga, Ge, and In within the Chipu Pb-Zn deposit and reveals the characteristics of the ore-forming fluid in the Chipu Pb-Zn deposit.

## 2. Regional Geology

Located in the western Yangtze Block, the SYG Pb-Zn metallogenic province serves as a transition area linking the Gondwana paleocontinent and the Laurasia paleocontinent, flanked by the Songpan-Ganzi Orogenic belts in the north, the Sanjiang Orogen belts in the southwest, and the Cathaysia Block in the southeast (Figure 1a) [25,26,28–30]. The area's strata are largely intact, predominantly consisting of the basement and sedimentary cover rocks. The Pb-Zn deposits in the region are predominantly found in carbonates from the Sinian to Permian periods [30–32]. The tectonic development in the area is remarkable, with its basement tectonics being E-W-trending and consisting of a series of faults and folds, while the cover is characterized by deep and large faults and retrograde fold tectonics, and the deposits are mainly distributed within the Xiaojiang Fault and its sub-fractures on the eastern side [25]. Four distinct faults, namely the Anninghe Fault, the Xiaojiang Fault, the Shizong-Shuicheng Fault, and the Ziyun-Yadu Fault, are intricately linked to the Pb-Zn mineralization (Figure 1b) [19]. These faults provide an important tectonic background for the mineralization and may serve as a conduit for the transport and enrichment of the mineral matter. The magmatic rocks in the region are characterized by both intrusive and extrusive types. During the Indosinian period, the area was marked by extensive development of the Emeishan basalts [14,30,33], with eruption ages ranging from approximately 261 Ma to 257 Ma [34].



**Figure 1.** (a) The SYG Pb-Zn metallogenic province in the Yangtze Block shown on a tectonic map of China (orange dashed rectangle) (adapted from [19]). Abbreviations: QT: Qiangtang terrane, LS: Lhasa terrane. (b) Regional geological map of the SYG Pb-Zn metallogenic province (adapted from [35,36]).

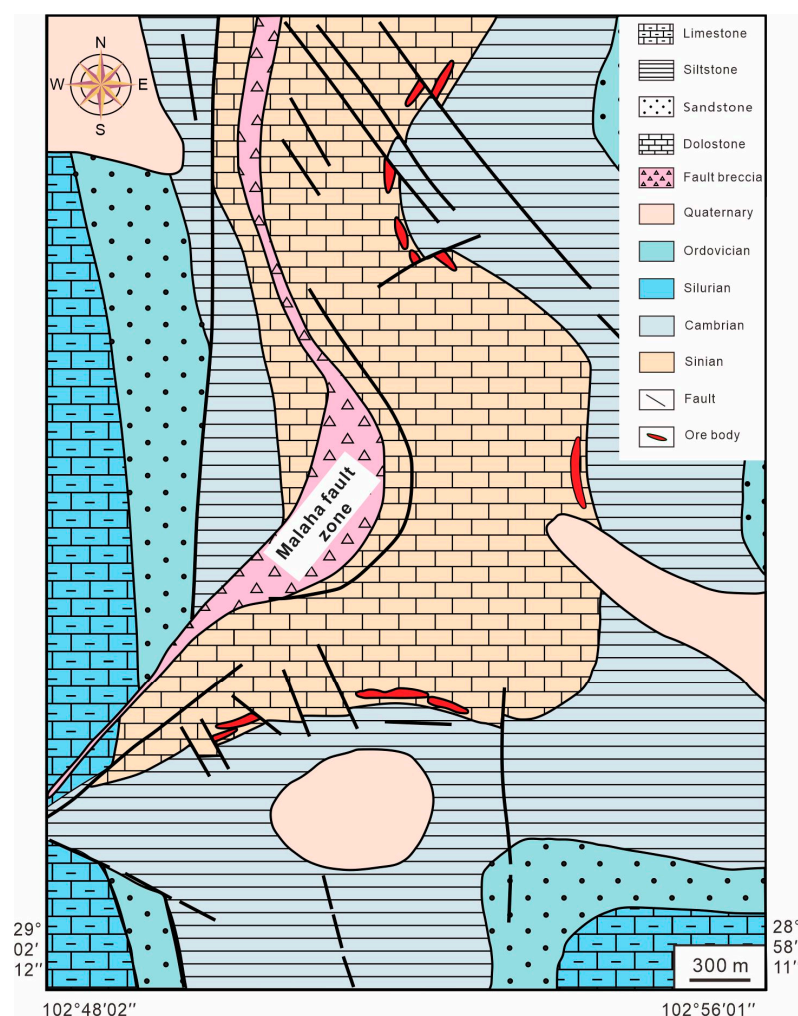
Over four hundred Pb-Zn deposits, including Huize, Tianbaoshan, Daliangzi, and Maliping, are found in the SYG Pb-Zn metallogenic region (Figure 1b) [30,36]. Known for their significant reserves and high-grade ores, these deposits account for as much as 30% of China's Pb and Zn resources [28,34,37]. For example, the Huize large Pb-Zn deposit has reserves of up to 5 Mt Pb + Zn, and the ore grade ranges from 25% to 30%. In addition, there are a lot of large-medium-small-sized Pb-Zn deposits with Pb + Zn reserves between 0.6 and 2.0 Mt, with ore grades between 8% and 10% [25,28,37].

### 3. Deposit Geology

The Chipu Pb-Zn deposit is located in Ganluo County, Sichuan Province. It is situated in the northern segment of the SYG Pb-Zn mineralization province and along the Ganluo River Fault, which is part of the eastern branch of the N-S-trending Xiaojiang Fault system [25,27,38]. The exposed strata are mainly composed of Sinian Dengying Formation dolomite, lower Cambrian Qiongzhusi Formation carbonaceous siltstone, middle to upper

Cambrian dolomite and dolomitic siltstone, Ordovician dolomite and sandstone, and Silurian siltstone intercalated with limestone. The contact between the Dengying Formation and the Cambrian System is characterized by an unconformity. A well-developed interlayer fracture zone exists between them, which provides favorable space for mineralization, with sphalerite and galena veins often filled within it.

The main structures with the mining region include the Malaha inverted anticline and the Malaha inverted fault, which trends in the NNW axial direction. The majority of ore bodies found within the mining region are situated in the dolomite of the Dengying Formation in the eastern flank of the Malaha inverted anticline, and most of them are stratiform and/or lenticular along the fracture zone (Figure 2) [25,27,38,39]. The occurrence of ore bodies was found to be basically consistent with that of the ore-bearing wall rock, with a dip of 10~50° [23,24]. Furthermore, the mining region features secondary faults that trend in the NW and NNW directions. These post-mineralization faults exert a certain degree of disruption on the ore body. The ore body contains Pb reserves of 0.3 Mt and Zn reserves of 0.12 Mt, with average grades of 7.4% and 3.1%, respectively, while the combined Pb + Zn grade can reach 10%~22% in certain areas [25,27,39].

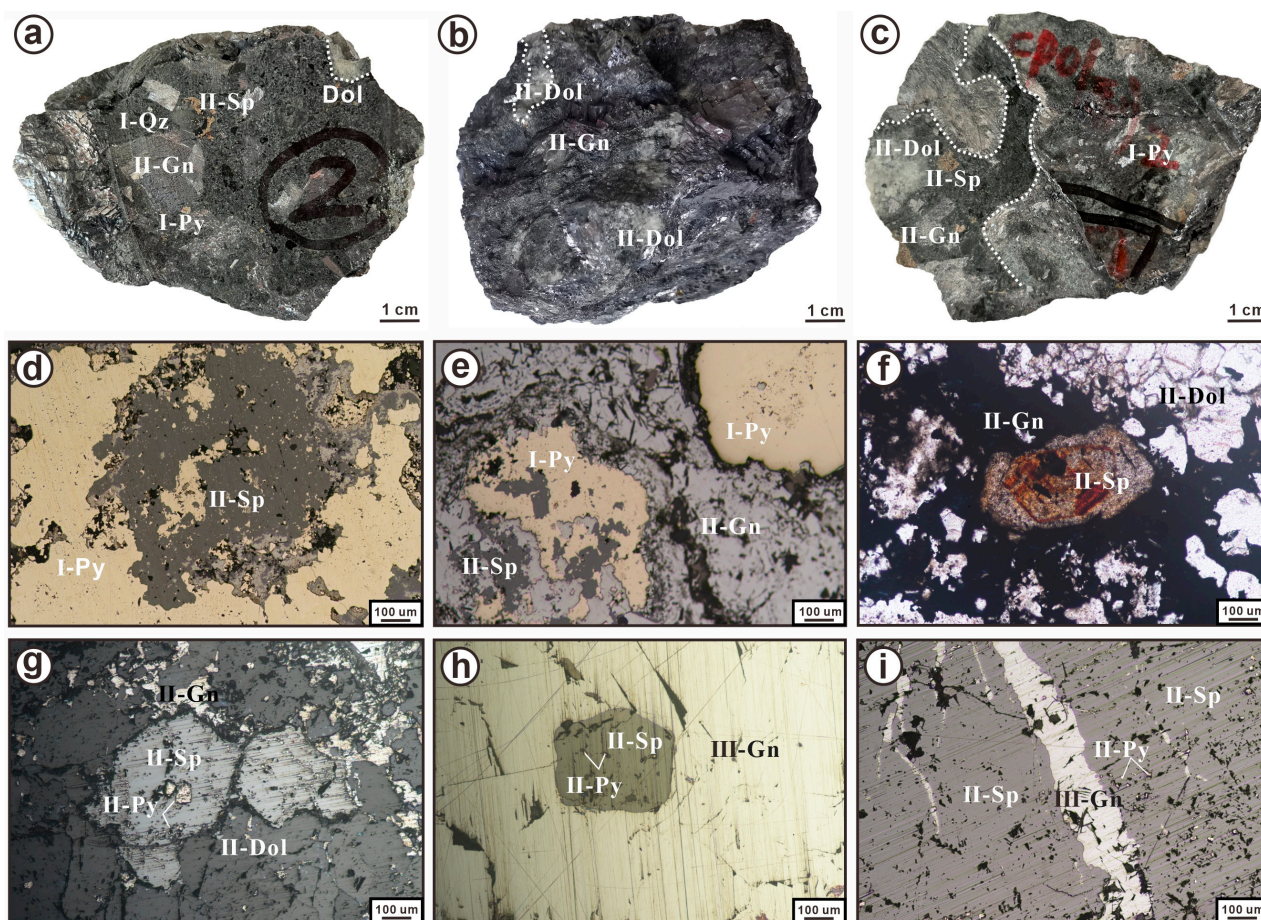


**Figure 2.** Geological map of the Chipu Pb-Zn deposit (adapted from [25,27]).

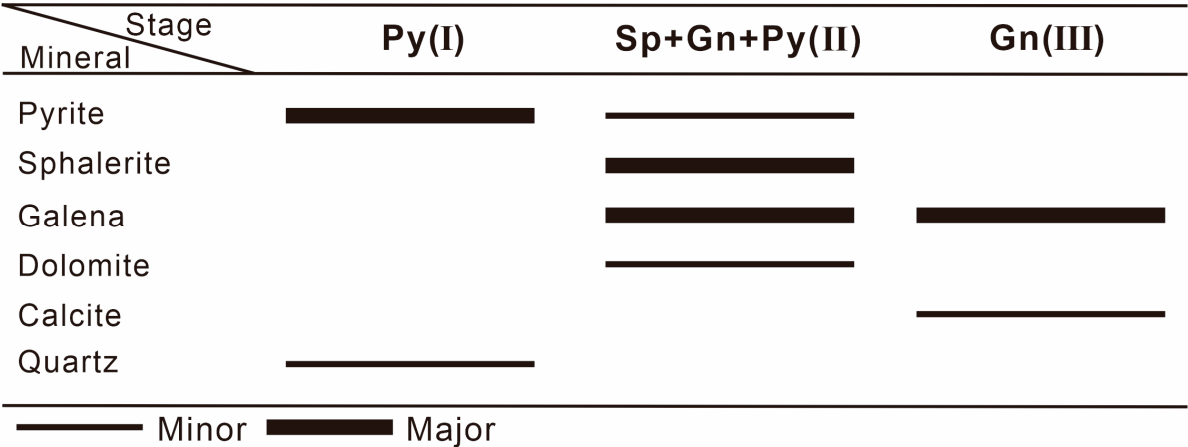
#### 4. Characteristics and Sequences of Mineralization

Pyrite, sphalerite, and galena dominate as the primary ore minerals in the deposit, with dolomite, quartz, and a minor amount of calcite being typical gangue minerals. Based on the paragenetic textures and mineral interconnections, the hydrothermal mineralization

period of the Chipu deposit is segmented into three stages: the pyrite stage (Stage I), the sphalerite–galena–pyrite stage (Stage II), and the galena stage (Stage III) (Figures 3 and 4).



**Figure 3.** Hand specimens and microscopic photographs show the main ore textures in the Chipu Pb-Zn deposit. (a) Pyrite (I-Py), quartz (I-Qz), galena (II-Gn), sphalerite (II-Sp), and dolomite (II-Dol) coexist in massive Pb-Zn ores. (b) Galena (II-Gn) and fine-vein dolomite (II-Dol) coexist in massive Pb-Zn ores. (c) Sphalerite (II-Sp), galena (II-Gn), and dolomite (II-Dol) coexist in sulfide veins. (d) Coarse-grained pyrite (I-Py) replaced by sphalerite (II-Sp) (reflected light). (e) Pyrite (I-Py) was replaced by galena (II-Gn) and sphalerite (II-Sp), presenting metasomatic-relict textures (reflected light). (f) Galena (II-Gn), sphalerite (II-Sp), and dolomite (II-Dol) coexist, and a sphalerite (II-Sp) rhythm zone can be seen (transmitted light). (g) Pentagonal sphalerite (II-Sp) is contained in dolomite, and fine-grained pyrite (II-Py) is distributed on the surface of sphalerite (II-Sp) and coexists with galena (II-Gn) (reflected light). (h) Granular sphalerite (II-Sp) is contained in galena (III-Gn), and stellate pyrite (II-Py) is distributed in sphalerite (II-Sp) (reflected light). (i) Coarse-grained sphalerite (II-Sp) cut by vein galena (II-Gn) and fine-grained pyrite (II-Py) distributed in sphalerite (II-Sp) (reflected light). Abbreviations: Gn—galena, Sp—sphalerite, Py—pyrite, Dol—dolomite.



**Figure 4.** Mineral paragenesis relationships of the Chipu Pb-Zn deposit. Abbreviations: Gn—galena, Sp—sphalerite, Py—pyrite.

During the pyrite stage (Stage I), the mineral assemblage is primarily composed of fine-grained quartz (I-Qz) and abundant coarse-grained, euhedral to subhedral pyrite (I-Py), which has experienced partial dissolution by later fluids or been replaced by subsequent sulfide minerals (Figure 3a,d,e). During the main ore stage, the sphalerite–galena–pyrite stage (Stage II), the mineral assemblage is primarily composed of fine to coarse-grained sphalerite (II-Sp), subhedral galena (II-Gn), dolomite (II-Dol), and fine-grained pyrite (II-Py) (Figure 3b,c,e–g). The coprecipitation of II-Dol, II-Sp, and II-Gn is distinctly marked by the presence of mutual growth boundaries (Figure 3f,g). It is noticeable that II-Py is dispersed across the surface of II-Sp or manifests as a fracture filling (Figure 3g–i). Furthermore, Figure 3f illustrates the frequent occurrence of rhythmic banding in I-Sp crystals, characterized by a distinct color alternation between light yellow and dark brown, which corresponds to fluctuations in trace element concentrations. The galena stage (Stage III) is characterized by anhedral or vein-like galena (III-Gn) (Figure 3h,i), in which the vein-like III-Gn cuts through the coarse-grained sphalerite, which hosts the fine-grained pyrite (Figure 3i). The detailed mineral paragenesis textures and mineral interconnections are shown in Figures 3 and 4.

5. Sampling and Analytical Methods

5.1. Sample Description

All samples in this study were collected from ore bodies in the Chipu No. 1 and No. 2 ore zone, with all specimens being sulfide-bearing ore samples. The sulfide’s structural and mineralogical properties were ascertained through an in-depth petrographic study of 18 polished sections and thin sections of typical lead-zinc ores from the Chipu deposit. Hence, the initial examination of these samples took place on a hand specimen scale, followed by the confirmation of mineral types and structures using the optical microscope, and culminating in the analysis of trace elements and in situ sulfur isotopes.

This research involved the detailed trace element analysis, yielding 57 points for sphalerite, 10 for pyrite, and 5 for single-mineral galena, as well as in situ sulfur isotope testing at 30 points for sphalerite, pyrite, and galena, all conducted on 18 lead-zinc ore specimens from the Chipu Pb-Zn deposit.

5.2. (LA)-ICP-MS Sulfides Trace Element Analysis

The trace element concentrations within sphalerite and pyrite were identified at Guangzhou Tuoyan Analytical Technology Co., Ltd., (Guangzhou, China), employing a NWR 193 nm ArF Excimer laser-ablation system coupled to an iCAP RQ (ICPMS). The

laser emitted a fluence of  $3.5 \text{ J/cm}^2$ , featuring a 6 Hz repetition rate, a spot size of  $30 \text{ }\mu\text{m}$ , and a 40 s analysis period, succeeded by a 40 s background evaluation. Two standard blocks (NIST 610 and GSE-2G) and a MASS-1 sulfide standard were succeeded by 5 to 8 unidentified samples. In the case of sphalerite and pyrite, measurements were taken of 21 isotopes, cumulatively amounting to a dwell time of 210 ms:  $^{34}\text{S}$ ,  $^{51}\text{V}$ ,  $^{53}\text{Cr}$ ,  $^{55}\text{Mn}$ ,  $^{57}\text{Fe}$ ,  $^{59}\text{Co}$ ,  $^{60}\text{Ni}$ ,  $^{65}\text{Cu}$ ,  $^{66}\text{Zn}$ ,  $^{71}\text{Ga}$ ,  $^{73}\text{Ge}$ ,  $^{75}\text{As}$ ,  $^{77}\text{Se}$ ,  $^{107}\text{Ag}$ ,  $^{111}\text{Cd}$ ,  $^{115}\text{In}$ ,  $^{118}\text{Sn}$ ,  $^{121}\text{Sb}$ ,  $^{126}\text{Te}$ ,  $^{205}\text{Tl}$ , and  $^{208}\text{Pb}$ . Calibration of  $^{34}\text{S}$ ,  $^{51}\text{V}$ ,  $^{53}\text{Cr}$ ,  $^{55}\text{Mn}$ ,  $^{57}\text{Fe}$ ,  $^{59}\text{Co}$ ,  $^{60}\text{Ni}$ ,  $^{65}\text{Cu}$ ,  $^{66}\text{Zn}$ ,  $^{71}\text{Ga}$ ,  $^{73}\text{Ge}$ ,  $^{75}\text{As}$ ,  $^{77}\text{Se}$ ,  $^{107}\text{Ag}$ ,  $^{111}\text{Cd}$ ,  $^{115}\text{In}$ ,  $^{118}\text{Sn}$ ,  $^{121}\text{Sb}$ ,  $^{126}\text{Te}$ ,  $^{205}\text{Tl}$ , and  $^{208}\text{Pb}$  primarily employed the MASS-1 sulfide standard (Fe = 15.6%) [40]. To prevent the mutual interference between In and Sn isotopes from affecting the test data during the analysis of sphalerite, the data were corrected following the method provided by [41]. The detailed correction process is described in Table S2. The corrected In concentrations were consistent with the measured In concentrations, confirming that the measured In concentrations reliably represent the actual In concentrations in sphalerite.

The preliminary treatment for the trace element testing of galena by ICP-MS was completed at Wuhan Shangpu Analytical Technology Co., Ltd. (Wuhan, China), and the experiment was conducted at the Analytical Testing Research Center of the Beijing Research Institute of Uranium Nuclear Geology (Beijing, China). The trace elements analyzed included Mn, Fe, Co, Ni, Cu, Ga, Ge, In, Cd, Tl, As, Se, Te, Ag, Sn, Sb, V, Cr, Zn, Pb, etc. The analysis began by selecting single-mineral galena and grinding it to 200 mesh, then placing the samples in a sealed digestion vessel, dissolving them with hydrofluoric acid and nitric acid, and heating on an electric hot plate to completely evaporate the hydrofluoric acid, followed by sealed dissolution with nitric acid. After appropriate dilution, the content of trace elements in the samples was directly determined using the external standard method of ICP-MS.

The trace element mapping of sphalerite were ascertained through a NWR 193 nm ArF Excimer laser-ablation system coupled to an iCAP RQ ICPMS at the Guangzhou Tuoyan Analytical Technology Co., Ltd. (Guangzhou, China). The examination of the sample included assessing 32 isotopes, noting the dwell times in milliseconds as follows, cumulatively amounting to a dwell time of 160 ms:  $^{27}\text{Al}$  (5),  $^{29}\text{Si}$  (5),  $^{39}\text{K}$  (5),  $^{43}\text{Ca}$  (5),  $^{57}\text{Fe}$  (5),  $^{71}\text{Ga}$  (5),  $^{73}\text{Ge}$  (5),  $^{82}\text{Se}$  (5),  $^{85}\text{Rb}$  (5),  $^{89}\text{Y}$  (5),  $^{91}\text{Zr}$  (5),  $^{93}\text{Nb}$  (5),  $^{111}\text{Cd}$  (5),  $^{118}\text{Sn}$  (5),  $^{126}\text{Te}$  (5),  $^{133}\text{Cs}$  (5),  $^{139}\text{La}$  (5),  $^{140}\text{Ce}$  (5),  $^{141}\text{Pr}$  (5),  $^{146}\text{Nd}$  (5),  $^{147}\text{Sm}$  (5),  $^{153}\text{Eu}$  (5),  $^{157}\text{Gd}$  (5),  $^{159}\text{Tb}$  (5),  $^{163}\text{Dy}$  (5),  $^{165}\text{Ho}$  (5),  $^{166}\text{Er}$  (5),  $^{169}\text{Tm}$  (5),  $^{172}\text{Yb}$  (5),  $^{175}\text{Lu}$  (5),  $^{178}\text{Hf}$  (5),  $^{181}\text{Ta}$  (5),  $^{182}\text{W}$  (5),  $^{185}\text{Re}$  (5), and  $^{205}\text{Tl}$  (5). The laser emitted a fluence of  $3.5 \text{ J/cm}^2$ , repeated at a rate of 25 Hz, and had a  $5 \text{ }\mu\text{m}$  spot size, equivalent to a scanning speed of  $15 \text{ }\mu\text{m/s}$ . Calibration of the trace elements was conducted using the NIST 610 as an external benchmark.

### 5.3. In Situ Sulfur Isotope Analysis

The in situ sulfur isotope analysis of sulfide was performed with LA-MC-ICP-MS at Guangzhou Tuoyan Analytical Technology Co., Ltd., (Guangzhou, China). The employed laser-stripping apparatus was a 193 nm excimer laser-stripping system (RESolution M50-LR, ASI), produced by ASI, Australia. The sulfur isotope testing was performed with a high-resolution multi-collector inductively coupled plasma mass spectrometer (Nu Plasma 1700 MC-ICP-MS) produced by Nu Instruments, UK. The laser exhibited an energy density (fluence) of  $3.6 \text{ J/cm}^2$ , a frequency of 3 Hz, a spot size ranging from 25 to  $37 \text{ }\mu\text{m}$ , and a single-point stripping mode. The mode used for gathering data was TRA mode, with an integration duration of 0.2 s, a background capture time of 30 s, a sample integration period of 50 s, and a purge duration of 75 s. Other parameters of the instrument were set as follows: RF power at 1300 W; Neb atomization gas injection at  $0.8 \text{ mL/min}$ ; plasma gas injection at  $13 \text{ L/min}$ . All the internal benchmarks of S isotopes were ascertained

using gas-stable isotope mass spectrometry (GSIMS) or MC-ICP-MS with solution injection, employing IAEA-S-1, IAEA-S-2, and IAEA-S-3 (Ag<sub>2</sub>S powder) as the benchmark samples for this analysis. The detailed analytical methods used are described in [42–44].

## 6. Results

### 6.1. Trace Element Compositions of Sulfides

The trace element compositions of sphalerite, pyrite, and galena in the Chipu deposit, including Fe, Mn, Cu, Ga, Ge, In, Cd, Ag, As, Sn, Sb, Tl, Bi, Rb, and Pb, with the specific findings are presented in Table 1 and Figure 5 (the raw data are listed in Table S1). Gallium is mainly found in sphalerite (Figure 5), which has a high level content of 17–420 ppm. In contrast, pyrite and galena show lower concentrations of Ga, with 0.16–0.49 ppm in galena, and the highest Ga content in pyrite is 2.88 ppm. Sphalerite contains the highest concentration of germanium, with concentrations between 3.87 and 444 ppm, potentially attributable to the irregular dispersal of Ge within the sphalerite. The Ge content in pyrite is 4.53–93 ppm, and that in galena is lower than the detection limit. The indium shows significantly higher concentrations in sphalerite, mainly between 31 and 720 ppm. The levels of In in galena and pyrite are notably low, with the highest content in galena being 0.50 ppm, with the majority of other elements being below the detection limit, while the highest indium content in pyrite is 5.16 ppm.

**Table 1.** Trace elements in sulfide minerals from the Chipu Pb–Zn deposit (ppm).

| Sphalerite | Mn  | Fe   | Co   | Cu   | Ga  | Ge   | As   | Ag   | Cd   | In  | Sn   | Sb   | Pb   |
|------------|-----|------|------|------|-----|------|------|------|------|-----|------|------|------|
| CP-01A-2   | 319 | 282  | 0.26 | 1045 | 66  | 123  | 434  | 30   | 1006 | 166 | 7.64 | 454  | 38   |
|            | 357 | 531  | 0.94 | 1881 | 84  | 213  | 111  | 37   | 992  | 182 | 8.79 | 895  | 76   |
|            | 352 | 463  | 0.80 | 1292 | 56  | 162  | 68   | 27   | 1034 | 205 | 5.70 | 531  | 45   |
| CP-01A-3   | 344 | 490  | -    | 1809 | 59  | 158  | 116  | 38   | 998  | 277 | 5.09 | 915  | 101  |
|            | 335 | 587  | -    | 1375 | 98  | 195  | 56   | 31   | 984  | 472 | 18   | 333  | 28   |
|            | 336 | 653  | 0.28 | 1573 | 155 | 395  | 22   | 19   | 959  | 552 | 24   | 59   | 38   |
|            | 342 | 859  | 1.88 | 1255 | 115 | 64   | 36   | 14   | 972  | 679 | 130  | 209  | 15   |
|            | 348 | 862  | 2.56 | 1804 | 223 | 60   | 89   | 18   | 984  | 720 | 90   | 588  | 75   |
|            | 348 | 695  | 2.30 | 2309 | 420 | 120  | 119  | 28   | 999  | 562 | 42   | 848  | 82   |
| CP-01A-4   | 372 | 528  | 2.30 | 779  | 41  | 15   | 50   | 14   | 1001 | 249 | 4.79 | 449  | 122  |
|            | 367 | 517  | 2.50 | 1352 | 29  | 50   | 109  | 21   | 967  | 260 | 0.95 | 855  | 220  |
| CP-01B-2   | 344 | 937  | 1.68 | 1438 | 88  | 100  | 113  | 31   | 899  | 31  | 1.45 | 1022 | 183  |
|            | 334 | 888  | 1.61 | 1940 | 17  | 154  | 137  | 21   | 886  | 48  | 1.53 | 1388 | 224  |
|            | 323 | 847  | 1.83 | 4711 | 171 | 444  | 399  | 57   | 861  | 90  | 1.77 | 3337 | 1005 |
|            | 326 | 730  | 1.20 | 467  | 75  | 78   | 8.51 | 5.07 | 894  | 107 | 14   | 69   | 14   |
|            | 319 | 557  | 1.24 | 706  | 175 | 71   | 5.99 | 5.53 | 929  | 176 | 51   | 68   | 2.43 |
| CP-02B-1   | 351 | 640  | 0.89 | 735  | 23  | 5.21 | 17   | 12   | 975  | 400 | 94   | 194  | 192  |
|            | 349 | 598  | 0.59 | 2841 | 85  | 48   | 165  | 43   | 985  | 452 | 121  | 1703 | 485  |
|            | 370 | 1154 | 0.71 | 6381 | 127 | 241  | 326  | 90   | 993  | 516 | 45   | 4087 | 889  |
|            | 351 | 561  | 0.36 | 2186 | 152 | 22   | 84   | 19   | 1020 | 400 | 9.54 | 1486 | 265  |
|            | 355 | 483  | 0.59 | 661  | 46  | 3.87 | 18   | 8.77 | 1145 | 280 | 83   | 158  | 84   |
| CP-03      | 96  | 2272 | 1.69 | 950  | 36  | 80   | 27   | 34   | 868  | 606 | 13   | 146  | 141  |
|            | 94  | 2251 | 1.73 | 1392 | 43  | 295  | 22   | 19   | 835  | 628 | 4.12 | 55   | 22   |
|            | 97  | 2200 | 1.63 | 798  | 32  | 17   | 28   | 23   | 879  | 618 | 24   | 117  | 12   |
|            | 105 | 2179 | 1.69 | 2154 | 132 | 121  | 109  | 52   | 935  | 541 | 109  | 759  | 43   |
|            | 109 | 2169 | 1.50 | 1019 | 100 | 51   | 27   | 15   | 995  | 406 | 146  | 184  | 8.45 |
|            | 98  | 2261 | 1.47 | 1475 | 25  | 325  | 44   | 37   | 851  | 514 | 3.75 | 141  | 18   |
|            | 100 | 2219 | 1.77 | 1343 | 43  | 278  | 27   | 25   | 869  | 593 | 30   | 85   | 15   |
|            | 104 | 2186 | 1.67 | 1313 | 30  | 142  | 83   | 44   | 873  | 610 | 47   | 301  | 59   |
|            | 109 | 2284 | 2.03 | 713  | 50  | 21   | 24   | 17   | 977  | 458 | 38   | 147  | 16   |
|            | 118 | 2160 | 0.95 | 1516 | 235 | 109  | 70   | 41   | 1079 | 297 | 59   | 466  | 28   |

Table 1. Cont.

| Pyrite   | Mn   | Fe     | Co   | Cu   | Ga   | Ge   | As   | Ag   | Cd   | In   | Sn   | Sb  | Pb    |
|----------|------|--------|------|------|------|------|------|------|------|------|------|-----|-------|
| CP-01A-2 | -    | 460000 | 21   | 344  | 0.49 | 93   | 909  | 3.15 | 4.35 | 5.16 | 33   | 85  | 23    |
| CP-01A-3 | 174  | 460000 | -    | 215  | 0.02 | 5.55 | 4269 | 8.88 | 0.47 | -    | -    | 190 | 280   |
| CP-01B-2 | 7.51 | 460000 | 68   | 1702 | 2.88 | 37   | 653  | 61   | 39   | 3.83 | 1.51 | 755 | 11383 |
| CP-02A   | 342  | 460000 | 7.36 | 183  | -    | 4.53 | 4409 | 40   | 1.81 | -    | -    | 335 | 1598  |
| CP-02B   | 12   | 460000 | 373  | 281  | 0.05 | 5.85 | 2743 | 31   | 0.59 | -    | -    | 122 | 1177  |
| CP-03    | -    | 460000 | 0.86 | 1468 | 0.69 | 12   | 446  | 125  | 0.34 | -    | -    | 833 | 828   |
| Galena   | Cu   | Co     | Ga   | Cd   | In   | Tl   | Re   | Sb   | Bi   | Rb   | V    | Pb  | W     |
| CP-01A   | 189  | 0.10   | 0.49 | 8.59 | 0.12 | 7.99 | -    | 1001 | 5.11 | 0.30 | 71   | /   | 0.13  |
| CP-01B   | 20   | 0.72   | 0.21 | 14   | 0.50 | 7.84 | -    | 771  | 4.99 | 0.10 | 5.73 | /   | 0.06  |
| CP-02A   | 85   | 0.21   | 0.16 | 6.34 | 0.15 | 7.97 | -    | 2800 | 5.28 | 0.12 | 26   | /   | 0.11  |
| CP-02B   | 53   | 0.14   | 0.40 | 8.28 | 0.19 | 7.27 | -    | 661  | 4.82 | 0.13 | 58   | /   | 0.05  |
| CP-03    | 40   | 0.07   | 0.48 | 13   | 0.05 | 9.42 | -    | 683  | 5.35 | 0.10 | 70   | /   | 0.05  |

“-” means concentration lower than detection limits; “/” means concentration higher than detection limits.

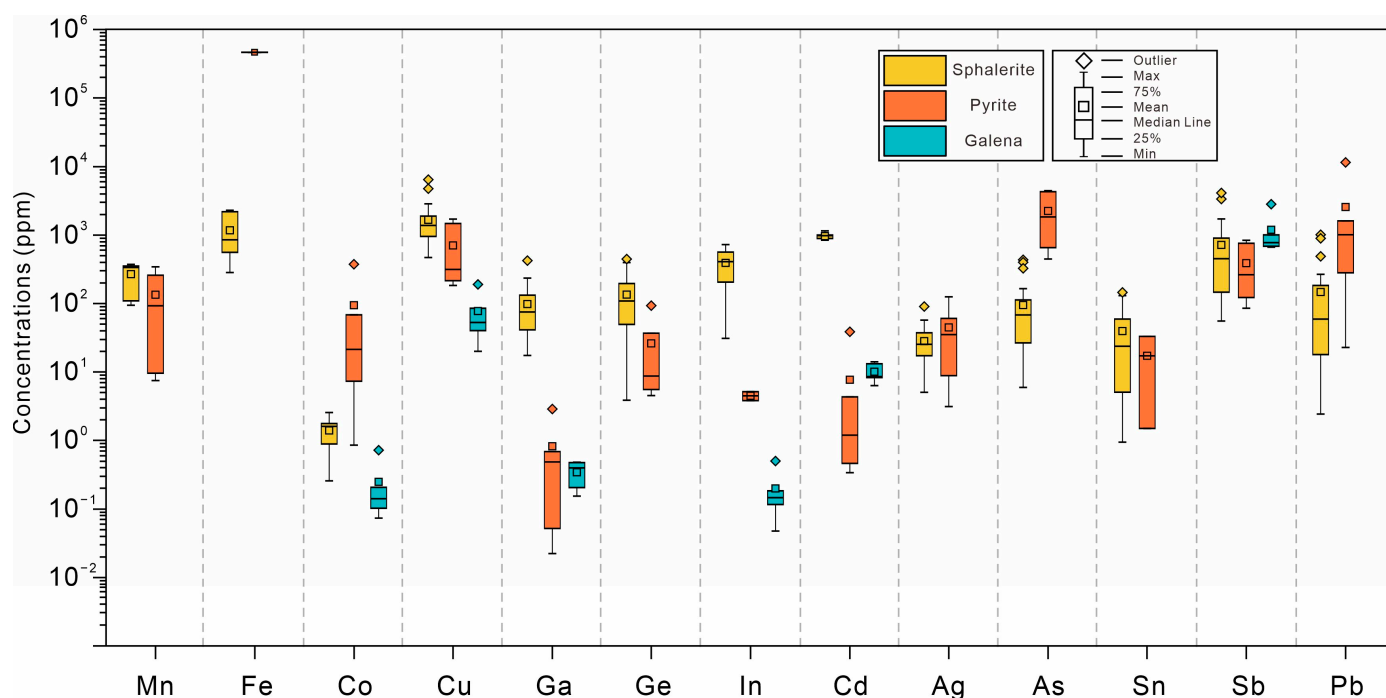


Figure 5. Box and whisker plots showing trace elements in sphalerite, pyrite, and galena from the Chipu Pb-Zn deposit.

Apart from the dispersed Ga, Ge, and In, the analytical examination further shows that the patterns of enrichment for other trace elements in sulfides are markedly distinct. Sphalerite is predominantly enriched with Mn, Cu, Cd, and Sn, with respective average concentrations of 272 ppm, 1670 ppm, 963 ppm, and 40 ppm. The pyrite is predominantly enriched with As, with an average content of 2238 ppm. Galena shows a relative enrichment in Sb, Tl, and V, with respective average concentrations of 1183 ppm, 8.10 ppm, and 46 ppm. In summary, sphalerite predominantly serves as the primary carrier for the dispersed Ga, Ge, and In in the Chipu deposit.

## 6.2. Sulfur Isotopic Compositions

The S isotope results for different sulfides in the Chipu deposit are listed in Table 2. Across twenty samples of sphalerite, the  $\delta^{34}\text{S}_{\text{V-CDT}}$  values span from +8.40‰ to +24.74‰, averaging at +17.37‰. The  $\delta^{34}\text{S}_{\text{V-CDT}}$  values of the five galena samples ranged from +13.23

to +16.37‰, averaging at +13.93‰. The  $\delta^{34}\text{S}_{\text{V-CDT}}$  values of the four pyrite samples ranged widely between +3.48 and +10.47‰, averaging +5.82‰ (Table 2).

**Table 2.** LA-MC-ICP-MS in situ S-isotope composition of various sulfides from the Chipu Pb-Zn deposit.

| Sample No. | Measured Mineral | $\delta^{34}\text{S}_{\text{V-CDT}}$ (‰) | 2SE  |
|------------|------------------|------------------------------------------|------|
| CP-01A-2   | sphalerite       | +24.13                                   | 0.18 |
| CP-01A-2   | sphalerite       | +23.83                                   | 0.19 |
| CP-01A-3   | sphalerite       | +24.74                                   | 0.18 |
| CP-02A-1   | sphalerite       | +24.31                                   | 0.18 |
| CP-02A-1   | sphalerite       | +23.06                                   | 0.21 |
| CP-01B-2   | sphalerite       | +19.64                                   | 0.20 |
| CP-02B-3   | sphalerite       | +17.72                                   | 0.21 |
| CP-01-1    | sphalerite       | +16.20                                   | 0.19 |
| CP-01-5    | sphalerite       | +17.50                                   | 0.24 |
| CP-02-3    | sphalerite       | +20.15                                   | 0.20 |
| CP-03-1    | sphalerite       | +24.11                                   | 0.18 |
| CP-03-1    | sphalerite       | +18.30                                   | 0.19 |
| CP-01A-3   | sphalerite       | +12.37                                   | 0.21 |
| CP-01B-2   | sphalerite       | +14.70                                   | 0.21 |
| CP-02B-2   | sphalerite       | +8.40                                    | 0.20 |
| CP-02B-2   | sphalerite       | +10.51                                   | 0.19 |
| CP-02B-3   | sphalerite       | +14.84                                   | 0.19 |
| CP-01-1    | sphalerite       | +13.02                                   | 0.20 |
| CP-02-2    | sphalerite       | +11.23                                   | 0.18 |
| CP-02-3    | sphalerite       | +8.66                                    | 0.85 |
| CP-01A-1   | galena           | +13.40                                   | 0.40 |
| CP-01B-1   | galena           | +13.36                                   | 0.27 |
| CP-02B-1   | galena           | +13.23                                   | 0.23 |
| CP-02-1    | galena           | +13.31                                   | 0.24 |
| CP-03-1    | galena           | +16.37                                   | 0.24 |
| CP-02B-1   | pyrite           | +10.47                                   | 0.23 |
| CP-01-5    | pyrite           | +3.48                                    | 0.25 |
| CP-02-3    | pyrite           | +4.26                                    | 0.24 |
| CP-02-3    | pyrite           | +5.09                                    | 0.23 |

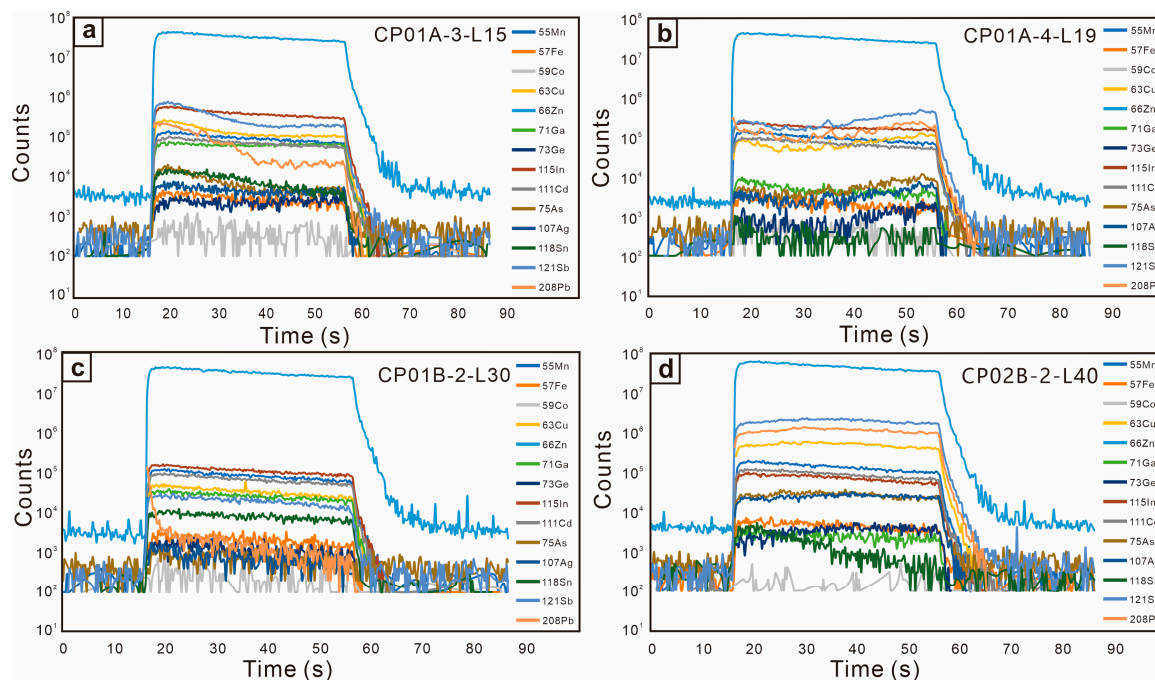
## 7. Discussion

### 7.1. Distributions and Substitutions of Trace Elements in Sphalerite

As mentioned above, the sphalerite in the Chipu deposit predominantly concentrates Ga, Ge, and In (Figure 5). The trace elements within sphalerite are predominantly present as solid solutions, manifesting as nanoparticles or mineral microinclusions [45,46]. The fluctuations in LA-ICP-MS signals from time-resolved depth profiles can be used to effectively determine the occurrence states of trace elements. Typically, a “flat” signal spectrum signifies either a solid solution or a uniformly distributed nanoparticle, whereas strong signal fluctuations always indicate an uneven distribution of mineral microinclusions or nanoparticles [17,45,47,48].

This research reveals that most trace elements, including Mn, Ga, Ge, In, Fe, Cu, As, Ag, Sn, Sb, Pb, and Cd, display a level profile (as shown in Figure 6), indicating their existence in solid solutions or uniformly distributed nanoparticles. The mapping image of a single sphalerite grain (Figure 7) reveals distinct zoning patterns among critical dispersed metals. Ga and Ge exhibit heterogeneous distributions, with their enrichment zones closely overlapping those of Cu and displaying concentric zoning patterns concentrated in the core and rim domains of the sphalerite, corresponding to its dark-brown colored regions. This spatial correspondence suggests a strong correlation among Ga, Ge, and Cu. In contrast, In

shows a relatively homogeneous distribution that aligns spatially with Fe-enriched regions, indicating a robust positive correlation between In and Fe.

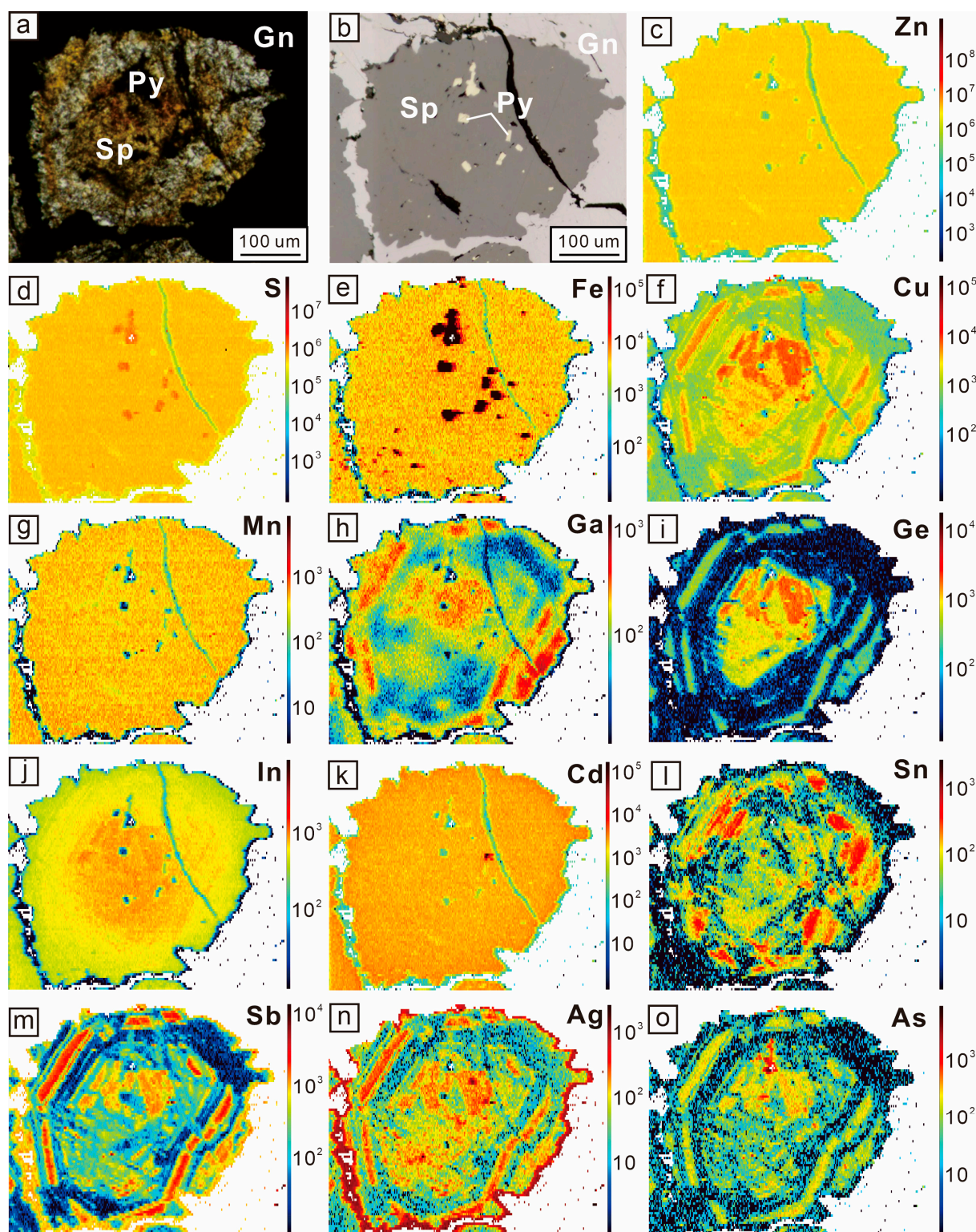


**Figure 6.** Representative single-spot LA-ICP-MS spectra for sphalerite from the Chipu Pb-Zn deposit.

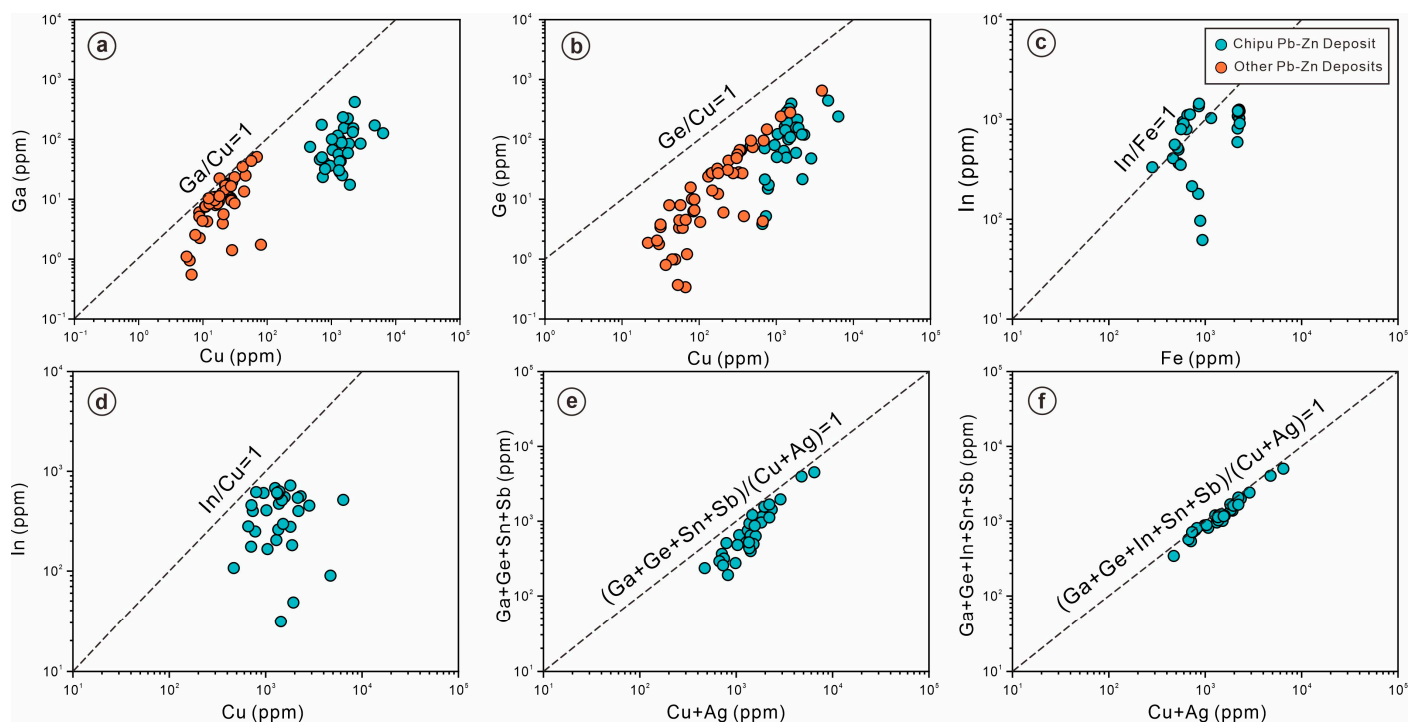
The frequent occurrence of coupling substitution in sphalerite is attributed to the similarity in the radii of  $\text{Ga}^{3+}$  (0.62 Å),  $\text{Ge}^{4+}$  (0.53 Å),  $\text{In}^{3+}$  (0.81 Å),  $\text{Fe}^{2+}$  (0.72 Å),  $\text{Cu}^+$  (0.96 Å),  $\text{Sn}^{4+}$  (0.71 Å), and  $\text{Sn}^{2+}$  (0.93 Å) with that of  $\text{Zn}^{2+}$  (0.74 Å) [11,44]. A variety of possible substitution mechanisms for the incorporation of Ga, Ge, and In into the sphalerite lattice have been proposed, such as  $\text{Ga}^{3+} + (\text{Ag}^+, \text{Cu}^+) \leftrightarrow 2\text{Zn}^{2+}$ ,  $\text{Ge}^{4+} + 2(\text{Ag}^+, \text{Cu}^+) \leftrightarrow 3\text{Zn}^{2+}$ ,  $\text{In}^{3+} + \text{Cu}^+ \leftrightarrow 2\text{Zn}^{2+}$ ,  $\text{In}^{3+} + \text{Sn}^{2+} + (\text{Ag}^+, \text{Cu}^+) \leftrightarrow 3\text{Zn}^{2+}$ , and  $\text{In}^{3+} + \text{Sn}^{4+} + (\text{Ag}^+, \text{Cu}^+) + \square \leftrightarrow 4\text{Zn}^{2+}$  [7,17,47,49–54]. In instances of copper-deficient sphalerite (0.29–19.20 ppm), Ge can be incorporated into the sphalerite through a coupled substitution with  $\text{Pb}^{2+}$  [55].

As shown in Figure 8a,b, the Chipu sphalerite exhibits elevated levels of Ga, Ge, and Cu relative to other deposits, along with  $\text{Ga}/\text{Cu} = 1$  and  $\text{Ge}/\text{Cu} = 1$ . This is akin to the coupled substitution of Ga and Ge with  $\text{Cu}^+$  observed in other deposits [56,57]. Based on the spatial correlations revealed by mapping images among Ga, Ge, and Cu, coupled with the “flat” signal of Ga and Ge observed in time-resolved depth profiles, we propose that these elements primarily incorporate into the sphalerite lattice via isomorphic substitution, indicative of the substitution mechanism involving  $\text{Ga}^{3+} + \text{Cu}^+ \leftrightarrow 2\text{Zn}^{2+}$  and  $\text{Ge}^{4+} + 2\text{Cu}^+ \leftrightarrow 2\text{Zn}^{2+}$ . Notably, the marked association between  $\text{Cu} + \text{Ag}$  and  $\text{Ga} + \text{Ge} + \text{Sn} + \text{Sb}$ , following an equality line, implies a probable integration of these elements into sphalerite via coupled substitution (Figure 8e).

In is strongly correlated with Fe (Figures 7 and 8c), and negatively correlated with Cu (Figure 8d), indicating that In enters the sphalerite lattice mainly through coupling with  $\text{Fe}^{2+}$  and that the coupling substitution mechanism probably occurs through  $2\text{In}^{3+} + \text{Fe}^{2+} \leftrightarrow 4\text{Zn}^{2+}$ . A significant positive correlation exists between  $\text{Cu} + \text{Ag}$  and  $\text{Ga} + \text{Ge} + \text{Sn} + \text{Sb}$  (Figure 8e), with the addition of In strengthens the relationship of element coupling at a ratio of  $(\text{Ga} + \text{Ge} + \text{In} + \text{Sn} + \text{Sb})/(\text{Cu} + \text{Ag}) = 1$  (Figure 8f), indicating that the multi-elemental coupling mechanism also promotes the entry of In into sphalerite.



**Figure 7.** (a) Transmitted light and (b) reflected light photomicrograph of sphalerite (sample CP-03A); (c–o) LA-ICP-MS trace element mapping images of sphalerite (sample CP-03A).



**Figure 8.** Diagrams of (a) Cu vs. Ga, (b) Cu vs. Ge, (c) Fe vs. In, (d) Cu vs. In, (e) Cu + Ag vs. Ga + Ge + Sn + Sb, and (f) Cu + Ag vs. Ga + Ge + In + Sn + Sb for sphalerite from the Chipu Pb-Zn deposit. The data for other deposits of the SYG Pb-Zn metallogenic province are from [56,57].

## 7.2. Precipitation Temperature of Sphalerite

The trace element composition of sphalerite is closely linked with its precipitation temperature [58–60]. Various geothermometry methods for sphalerite have been proposed [59–61]. In this research, taking into account the precision and applicability of various geothermometers, the geothermometry developed by [60] was employed to determine the precipitation temperature of sphalerite in the Chipu deposit. The calculation method of geological thermometers is proposed as the following equation:

$$T(^{\circ}\text{C}) = -(54.4 \pm 7.3) \cdot \text{PC1}^* + (208 \pm 10) \quad (1)$$

$$\text{PC1}^* = \ln \left[ \left( C_{\text{Ga}}^{0.22} \cdot C_{\text{Ge}}^{0.22} \right) / \left( C_{\text{Fe}}^{0.37} \cdot C_{\text{Mn}}^{0.20} \cdot C_{\text{In}}^{0.11} \right) \right] \quad (2)$$

in which  $C_{\text{Ga}}$ ,  $C_{\text{Ge}}$ ,  $C_{\text{In}}$ , and  $C_{\text{Mn}}$  are given in ppm, and  $C_{\text{Fe}}$  is given in wt.%. The results are shown in Table 3.

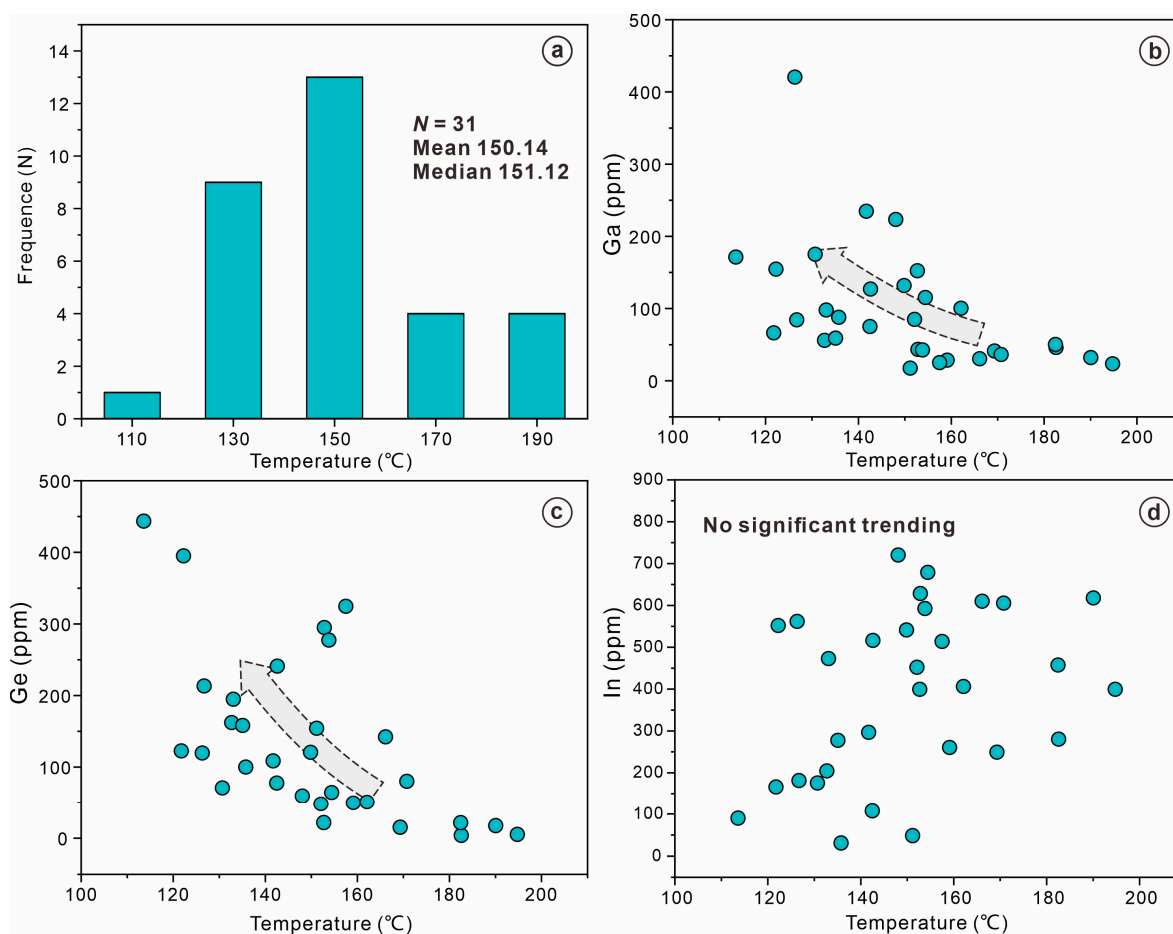
**Table 3.** Calculation results of the sphalerite mineralization temperature of the Chipu Pb-Zn deposit.

| Sample   | PC1* | T <sub>1</sub> /°C | T <sub>2</sub> /°C | T <sub>3</sub> /°C | T <sub>4</sub> /°C | T <sub>max</sub> /°C | T <sub>min</sub> /°C | T <sub>mean</sub> /°C |
|----------|------|--------------------|--------------------|--------------------|--------------------|----------------------|----------------------|-----------------------|
| CP-01A-2 | 1.6  | 120                | 143                | 100                | 123                | 143                  | 100                  | 122                   |
|          | 1.5  | 126                | 148                | 106                | 128                | 148                  | 106                  | 127                   |
|          | 1.4  | 133                | 153                | 113                | 133                | 153                  | 113                  | 133                   |
| CP-01A-3 | 1.3  | 135                | 155                | 115                | 135                | 155                  | 115                  | 135                   |
|          | 1.4  | 133                | 153                | 113                | 133                | 153                  | 113                  | 133                   |
|          | 1.6  | 121                | 144                | 101                | 124                | 144                  | 101                  | 122                   |
|          | 1.0  | 157                | 172                | 137                | 152                | 172                  | 137                  | 154                   |
|          | 1.1  | 150                | 166                | 130                | 146                | 166                  | 130                  | 148                   |
|          | 1.5  | 125                | 147                | 105                | 127                | 147                  | 105                  | 126                   |

Table 3. Cont.

| Sample   | PC1* | T <sub>1</sub> /°C | T <sub>2</sub> /°C | T <sub>3</sub> /°C | T <sub>4</sub> /°C | T <sub>max</sub> /°C | T <sub>min</sub> /°C | T <sub>mean</sub> /°C |
|----------|------|--------------------|--------------------|--------------------|--------------------|----------------------|----------------------|-----------------------|
| CP-01A-4 | 0.7  | 174                | 184                | 154                | 164                | 184                  | 154                  | 169                   |
|          | 0.9  | 163                | 176                | 143                | 156                | 176                  | 143                  | 159                   |
| CP-01B-2 | 1.3  | 136                | 155                | 116                | 135                | 155                  | 116                  | 136                   |
|          | 1.0  | 154                | 169                | 134                | 149                | 169                  | 134                  | 151                   |
|          | 1.7  | 111                | 136                | 91                 | 116                | 136                  | 91                   | 114                   |
|          | 1.2  | 144                | 161                | 124                | 141                | 161                  | 124                  | 142                   |
|          | 1.4  | 130                | 151                | 110                | 131                | 151                  | 110                  | 131                   |
| CP-02B-1 | 0.2  | 203                | 207                | 183                | 187                | 207                  | 183                  | 195                   |
|          | 1.0  | 155                | 170                | 135                | 150                | 170                  | 135                  | 152                   |
|          | 1.2  | 144                | 161                | 124                | 141                | 161                  | 124                  | 143                   |
|          | 1.0  | 155                | 170                | 135                | 150                | 170                  | 135                  | 153                   |
|          | 0.5  | 189                | 196                | 169                | 176                | 196                  | 169                  | 183                   |
| CP-03    | 0.7  | 176                | 186                | 156                | 166                | 186                  | 156                  | 171                   |
|          | 1.0  | 155                | 170                | 135                | 150                | 170                  | 135                  | 153                   |
|          | 0.3  | 198                | 202                | 178                | 182                | 202                  | 178                  | 190                   |
|          | 1.1  | 152                | 168                | 132                | 148                | 168                  | 132                  | 150                   |
|          | 0.8  | 166                | 178                | 146                | 158                | 178                  | 146                  | 162                   |
|          | 0.9  | 161                | 174                | 141                | 154                | 174                  | 141                  | 158                   |
|          | 1.0  | 157                | 171                | 137                | 151                | 171                  | 137                  | 154                   |
|          | 0.8  | 170                | 182                | 150                | 162                | 182                  | 150                  | 166                   |
|          | 0.5  | 189                | 196                | 169                | 176                | 196                  | 169                  | 182                   |
|          | 1.2  | 143                | 161                | 123                | 141                | 161                  | 123                  | 142                   |

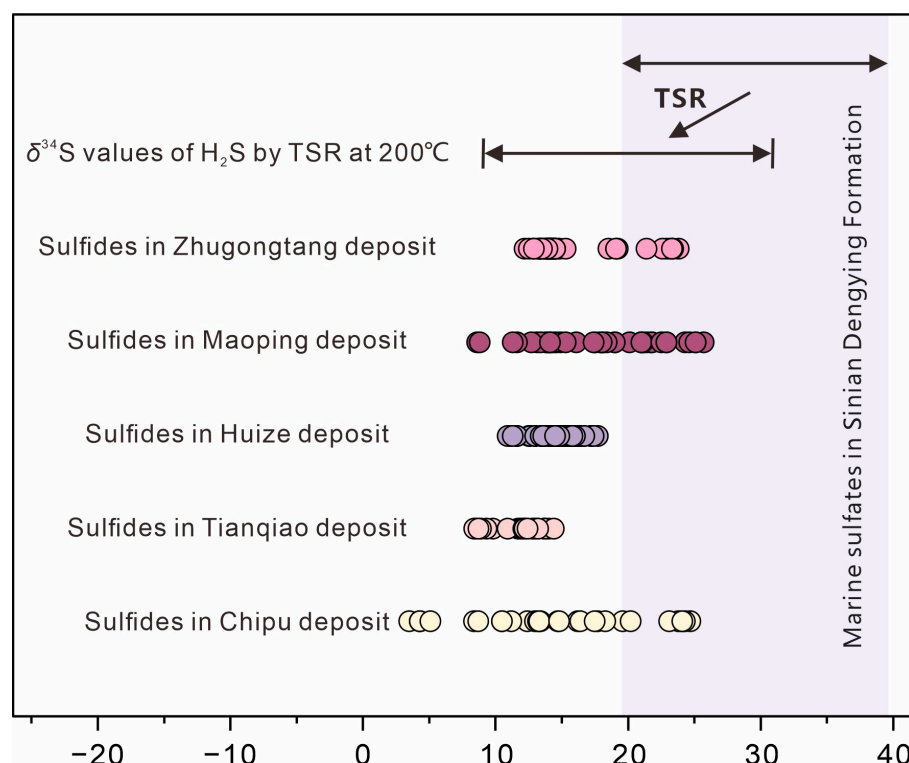
The sphalerite mineralization temperatures range from 114 °C to 195 °C (Table 3; Figure 9a), consistent with the homogenization temperature range (109–248 °C) measured by fluid inclusions within quartz in previous studies (109–248 °C) [22]. In summary, the mineralization of the deposit generally occurred under medium to low temperatures. Furthermore, as temperature decreases, the concentrations of Ga and Ge in sphalerite show varying degrees of increase, while the concentration of In exhibits no discernible trend with temperature variation (Figure 9b–d). Temperature serves as a critical factor controlling the enrichment of Ga and Ge in sphalerite. Although the detailed processes governing the enrichment of these scattered elements (Ga and Ge) in sphalerite remain unclear, numerous studies have demonstrated that their enrichment (100–200 °C) is closely associated with medium- to low-temperature mineralizing fluids [7,50]. In the Sichuan-Yunnan-Guizhou region, the Late Permian Emeishan mantle plume magmatic activity generated extensively distributed flood basalts with thicknesses ranging from hundreds to thousands of meters, along with substantial accumulations of mafic magmatic residues retained in the deep crust [62,63]. The Emeishan basalts exhibit elevated Ge ( $1.0 \times 10^{-6}$ – $2.3 \times 10^{-6}$ ) and Ga ( $20 \times 10^{-6}$ – $25 \times 10^{-6}$ ) contents [62,64]. These elements (Ge and Ga) within the basalts and deep crustal mafic magmatic residues could be leached by hydrothermal activities during metamorphism and alteration processes. Consequently, the Ga and Ge in sphalerite from the Chipu deposit likely originated from the Emeishan basalts, with the enrichment of these elements in sphalerite resulting from the extraction of source layers by mineralizing fluids (basinal brines).



**Figure 9.** (a) Temperature–frequency histogram of rim–core of sphalerite; (b–d) temperature versus Ga, Ge, and In contents of sphalerite from the Chipu Pb–Zn deposit. Temperature data were calculated from the LA–ICP–MS data of sphalerite [60].

### 7.3. Sources of Sulfur

In the Chipu deposit, the predominant sulfides are sphalerite, pyrite, and galena (Figure 3), with a lack of sulfate minerals. Consequently, the sulfur isotopic signatures of sulfides can be used to approximate the isotopic compositions of the ore-forming fluid [65,66]. In the Chipu deposit, the sulfide  $\delta^{34}\text{S}$  measurements during the ore-stage range between +3.48‰ and +24.74‰, with an overall average of approximately +15.19‰ for all the sulfides. In the SYG metallogenic province, the reduced sulfur in the majority of Pb–Zn deposits is typically believed to originate from marine sulfates (e.g., Huize; Tianqiao; Maoping; Zhugongtan) [36,56,67,68]. Several processes can convert marine sulfates into  $\text{H}_2\text{S}$ , predominantly through bacterial sulfate reduction (BSR) and thermochemical sulfate reduction (TSR) [66]. BSR is mainly characterized by large negative  $\delta^{34}\text{S}$  values [69], and TSR is typically characterized by positive  $\delta^{34}\text{S}$  values [70]. Moreover, all  $\delta^{34}\text{S}$  measurements of the sulfate in the Chipu deposit are positive and the formation temperature of the sphalerite ranges 122.25 to 206.55 °C, which is more favorable for TSR (commonly >130 °C) [71–73] than for BSR (generally 50–70 °C, with a maximum of 110 °C) [71,74]. Consequently, we infer that the reduced S in the Chipu deposit is sourced from marine sulfates in the mineral-bearing carbonates, through the process of TSR (Figure 10).

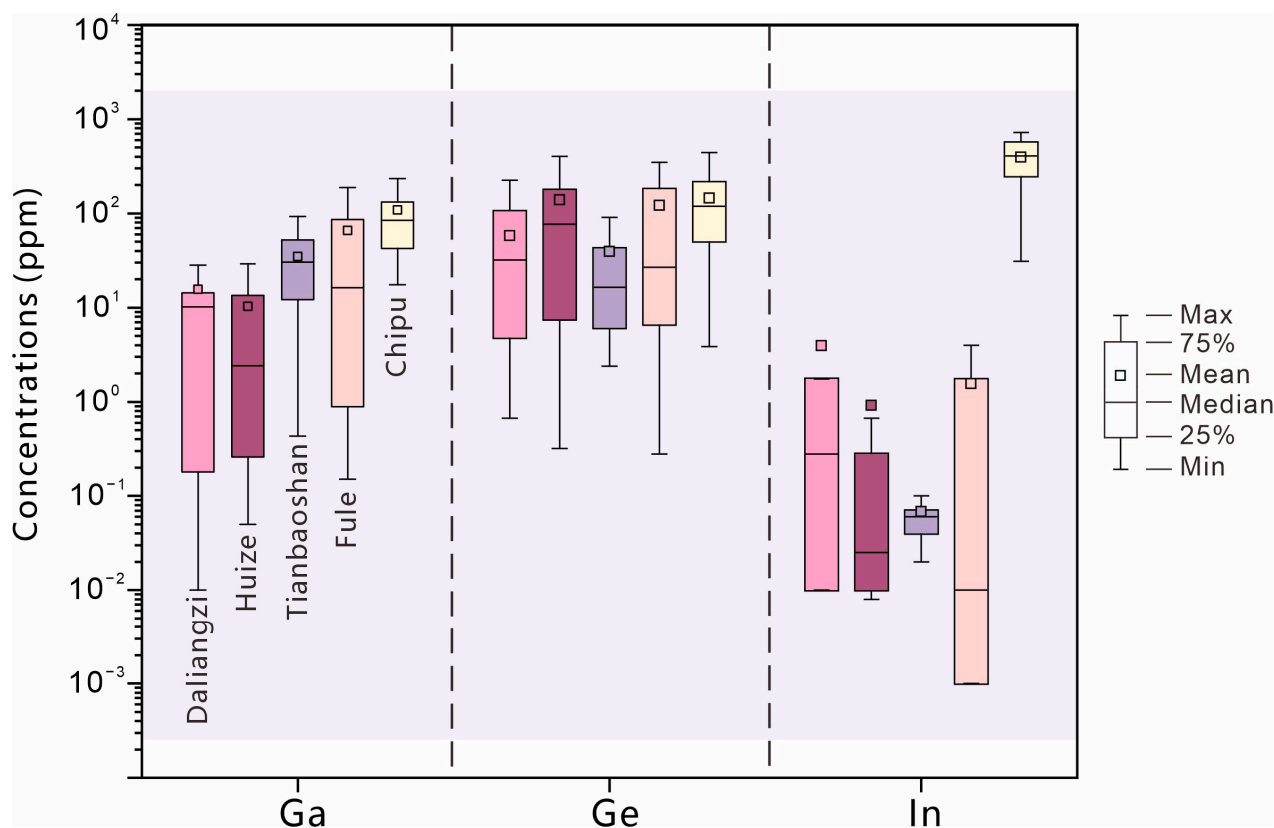


**Figure 10.** The  $\delta^{34}\text{S}$  of sulfides from the Chipu Pb-Zn deposit and other Pb-Zn deposits in the SYG metallogenic province. The  $\delta^{34}\text{S}$  values of marine sulfate in the Sinian Dengying Formation are from [75] (purple area); the  $\delta^{34}\text{S}$  values of other Pb-Zn deposits are from [36,56,67,68].

It has been reported that the  $\delta^{34}\text{S}$  measurements of marine sulfates within the Dengying Formation range from +20.2‰ to +38.7‰ (Figure 10) [75]. It is hypothesized that the TSR process involving reduced sulfur and marine sulfates results in sulfur isotope fractionation of about 20‰ at 100 °C, 15‰ at 150 °C, and 10‰ at 200 °C [71]. Through TSR at 200 °C, which corresponds to the mean temperature of the ore-forming fluid, marine sulfates are expected to yield reduced sulfur with  $\delta^{34}\text{S}$  values between +10‰ and +30‰, aligning with the S isotopic composition of the Chipu sulfides (Figure 10). Consequently, the sulfur in the Chipu deposit's sulfides is believed to have originated from marine sulfates in the Dengying Formation, with reduced sulfur from TSR potentially contributing to the mineralization process and the formation of metal sulfides.

#### 7.4. The Extraordinary Indium Enrichment in Sphalerite and Exploration Potential

Our study revealed that the sphalerite found in the Chipu deposit contains Ga and Ge levels akin to those in other deposits. However, the In content in the Chipu sphalerite is exceptionally high, ranging between 31 and 720 ppm (averaging 390 ppm), significantly surpassing the levels found in other Pb-Zn deposits within the SYG Pb-Zn metallogenic province (e.g., Daliangzi; Huize; Maoping; Tianbaoshan) (Figure 11) [17,19,20,76]. The Pb-Zn ore body of this deposit is found in the dolomite stratum of the Dengying Formation of the Upper Sinian Series, close to the basement stratum, and is obviously controlled by the local geological structure. In addition, some studies have reported that intermediate-felsic igneous rocks are easily enriched in In, with In content reaching up to 0.26 ppm [4,18].

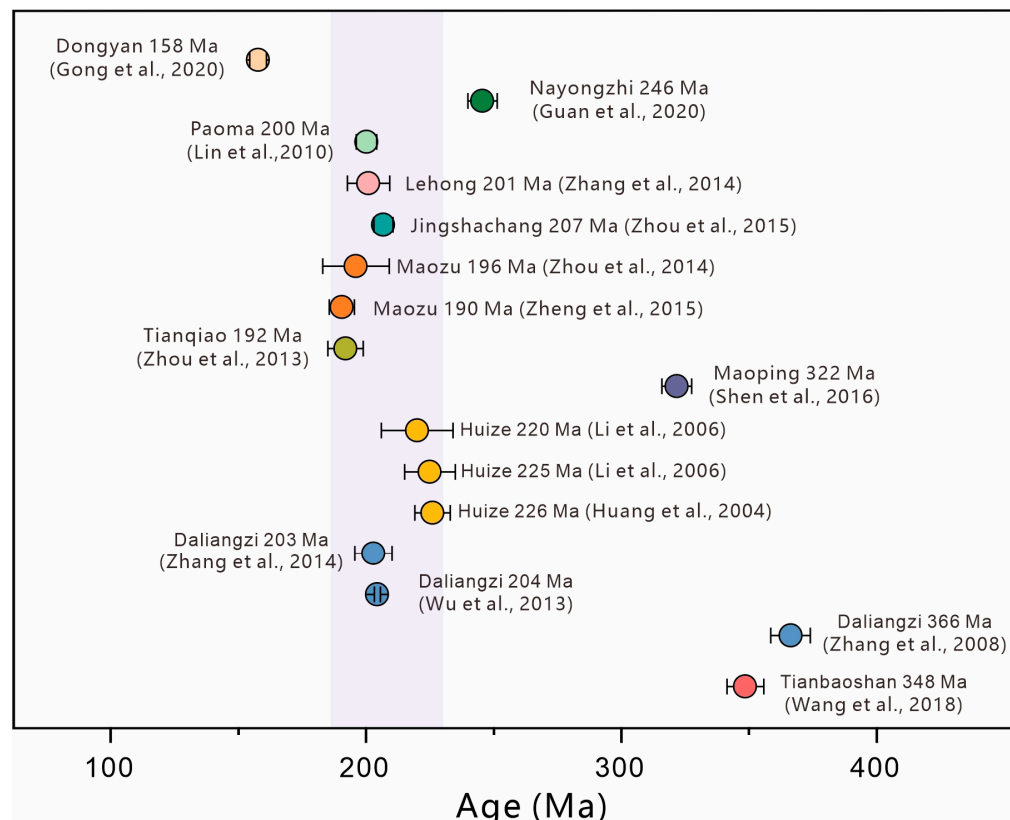


**Figure 11.** Comparison of Ga, Ge, and In contents in sphalerite between the Chipu deposit and other Pb-Zn deposit in the SYG metallogenic province. The data of other deposits are from [17,19,20,76].

Indium is predominantly hosted by sphalerite, which constitutes 95% of the global indium reserves [8]. The three main factors controlling the entry of In into sphalerite are as follows: (1) the ionic radius and similar ionic radius, which are more conducive to coupling substitution between elements [11]; (2) the ore-forming fluid temperature, where an increase in temperature enhances In enrichment [77]; (3) the enrichment of elements in sphalerite.

The main mineralization ages based on sphalerite and calcite dating in the Pb-Zn deposits along the western margin of the Yangtze Block range from 232 to 192 Ma (Figure 12). The Emeishan large igneous province is situated on the western margin of the Yangtze Block and features a large amount of Emeishan basalt, which is a product of mantle plume magmatic activity that occurred between 261 and 257 Ma [33,34]. The mineralization period of the Chipu Pb-Zn deposit is estimated to be between 230 and 200 Ma [78], which is subsequent to the mantle plume magmatic activity linked to the Emeishan large igneous province. However, paleomagnetic dating of six major Pb-Zn deposits across North America has revealed that Pb-Zn mineralization could span a duration of up to 25 Ma [5]. Thermal simulation studies on the evolution of the underplated Emeishan basalt indicate that the release of metal-containing fluid started to release 30 Ma post-underplating initiation, aligning with the age records of the Chipu deposit in this area. The simulation studies on the evolution of Emeishan basalt underplating suggest that the release of ore-forming fluid from the crystallizing underplated basalts can last for more than 100 Ma, encompassing nearly the entire range of SYG Pb-Zn mineralization age data [79]. Additionally, the reported metallogenic depth of the Chipu Pb-Zn deposit primarily ranges from 0.54 to 1.42 km [39]. If only the geothermal gradient (30 °C/km) is considered as the thermal source for metallogenesis, the theoretical mineralization temperature would be far below 100 °C, which is significantly lower than the actual mineralization temperature

range (114–195 °C). Therefore, the Emeishan mantle plume activity likely served as the primary thermal energy source for mineralization, providing the majority of the thermal energy required for basinal brine circulation. Thus, the Pb–Zn mineralization along the western margin of the Yangtze Block is likely associated with the mantle plume magmatic activity that characterized the Emeishan large igneous province [33]. Furthermore, a mantle upwelling in the Emeishan large igneous province led to crustal stretching and increased heat flow. This deep mantle activity also stimulated fluid circulation in the shallow crust, which in turn facilitated the formation of the Chipu deposit. Like the Huayuan Zn–Pb orefield, its origin is associated with the Cambrian mantle upwelling that occurred in the northeast region of Gondwana [80].



**Figure 12.** Age distribution histogram based on sphalerite and calcite dating of Pb–Zn deposits on the western margin of the Yangtze Block. The geochronology data of other Pb–Zn deposits are from [26,30,36,81–93].

As mentioned above, the magmatic activity linked to the Emeishan mantle plume could have facilitated the circulation of basinal brines, which in turn may have led to the extraction of In from the intermediate-felsic igneous rocks of the metamorphic basement. The increased presence of In in sphalerite is due to the similarity in geochemical properties between In and Zn, where elevated temperatures also enhance the synergistic incorporation of Fe and In into sphalerite. The In-enriched basinal brines, transported upward along the deep Malaha fault, accumulated in the high-porosity and high-permeability dolomite strata of the Dengying Formation, leading to the formation of the Chipu deposit with an abnormal enrichment of dispersed In. Furthermore, this study has positive implications for the exploration potential of Pb–Zn deposits with the same high In content characteristics on the western margin of the Yangtze Block.

## 8. Conclusions

- (1) Within the Chipu Pb-Zn deposit, sphalerite is mainly enriched in Mn, Cu, Ga, Ge, In, Cd, and Sn. Pyrite is mainly enriched in As. Galena is relatively enriched in Sb, Tl, and V. The dispersed Ga, Ge, and In are primarily enriched in sphalerite, and the substitution mechanisms include  $\text{Ga}^{3+} + \text{Cu}^+ \leftrightarrow 2\text{Zn}^{2+}$ ,  $\text{Ge}^{4+} + 2\text{Cu}^+ \leftrightarrow 3\text{Zn}^{2+}$ , and  $2\text{In}^{3+} + \text{Fe}^{2+} \leftrightarrow 4\text{Zn}^{2+}$ .
- (2) Geological features, geochemical studies, and sphalerite geothermometry results reveal that the mineralization processes in the Chipu deposit occurred within a range of medium to low temperatures (114–195 °C).
- (3) In situ sulfur isotopes from sphalerite, pyrite, and galena in the Chipu Pb-Zn deposit exhibit  $\delta^{34}\text{S}$  values ranging from +3.48 to +24.74‰, suggesting that marine sulfate present in the ore-bearing strata of the Dengying Formation is the predominant source of sulfur for the Chipu deposit.
- (4) The abnormal accumulation of In in sphalerite from the Chipu deposit may be related to the circulation of basinal brine promoted by the magmatic activity associated with the Emeishan mantle plume, and the basinal brine extracted significant amounts of In from the intermediate-felsic igneous rocks in the metamorphic basement.

**Supplementary Materials:** The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/min15040341/s1>, Table S1: Raw data for trace elements of sulfides by (LA)-ICP-MS; Table S2: calculation process for In contents correction of sphalerite by LA-ICP-MS; Table S3: Raw data of Figure 6; Table S4: Raw data of Figure 8; Table S5: Raw data of Figure 11; Table S6: Raw data of Figure 12.

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